



Frank Mohn Services AS

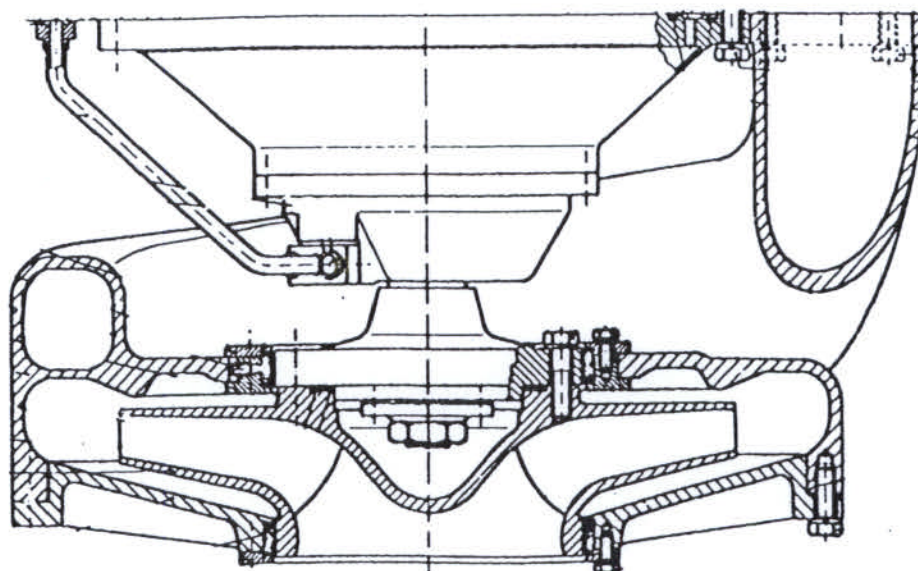
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Service Bulletin

MAKE YOU UP TO DATE WITH THE LATEST NEWS FROM FRANK MOHN SERVICES AS

Recondition of impellers & oversized wear-rings

Limitation of maintenance costs
of FRAMO cargo pumps



FRANK MOHN
SERVICES AS
Hardangerveien 150
P.O. Box 98
N-5051 Nesttun
Norway
Phone: 47 5 99 92 00
Telex: 42 927 framo n
Telefax: 47 5 99 93 82

FRANK MOHN
NEDERLAND BV
Edisonweg 14
P.O. Box 305
NL-3200 AH Spijkenisse
The Netherlands
Phone: 31 1880 19 311
Telex: 29 643 framo nl
Telefax: 31 1880 11 193

FRANK MOHN
HOUSTON INC
1802 West D-Street
P.O. Box 1586, La Porte
Texas 77571
USA
Phone: 1 713 471 7920
Telex: 68 68 120 framo lapl
Telefax: 1 713 470 1258

FRANK MOHN
SINGAPORE PTE LTD
16 Tuas Crescent
Singapore 2263
Republic of Singapore
Phone: 65 861 4455
Telex: 35 446 framo rs
Telefax: 65 861 5910

DEAR CUSTOMER !

INTRODUCTION

The main reason for introducing oversized wear-rings is to reduce the maintenance cost and improve the cargo pumping efficiency for your FRAMO Cargo Pumping System. This will increase the lifetime of the impeller and impeller flange and you will only have to renew the wear rings. We know that many ships today have the knowledge and equipment onboard that is necessary to perform this job.

However, should the vessel not be equipped for remachining and doing the job we would remind you that this reconditioning can of course be carried out by all FRAMO Service offices - in Rotterdam, Houston, Bergen and Singapore.

Worn-out wear rings

Worn wear rings has the following negative effects on the cargo pump performance:

- Loss of capacity
- Loss of efficiency
- Poor stripping
- Unstability when pumping in parallel
- Jet-erosion in pump sump

Lifetime of wear-rings and impeller as well as impeller flange depends upon the cleanness of the cargo being handled. Cargoes with a high number of impurities will naturally give a shortened lifetime to the wear rings as well as to the impeller and impeller flange.

When the gap between impeller and wear ring increases to more than 2-3 mm, we recommend that the rings are changed. Running the cargo pumps against closed cargo valve will normally give you a good picture of how the wear rings condition is.

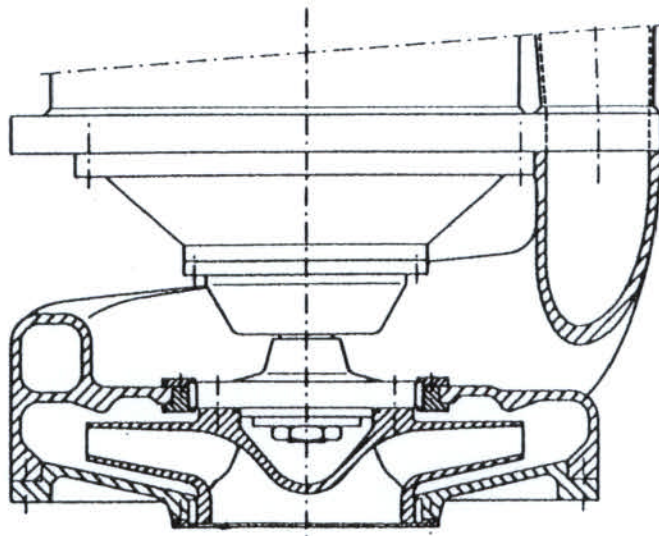
Normally you can expect to replace the wear rings with original size wear rings quite a few times before there is noticeable wear also on the impeller and impeller flange.

By introducing oversized wear rings you will be able to recondition the impeller and impeller flange, allowing further use of the existing impeller and impeller flange by using oversized wear rings.

So this will be a three step operation:

Step No. 1

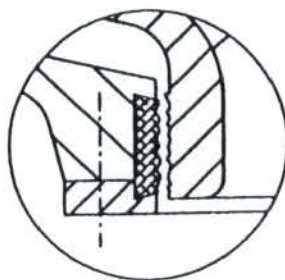
Original size wear rings



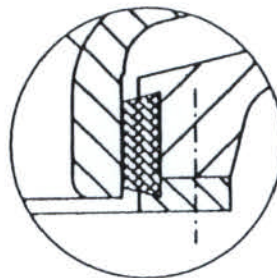
Original size wear rings can normally be installed quite a few times before the wear on the impeller and impeller flange increases to an intolerable level.

Step No. 2.

Oversized wear rings



Worn-out original
wear ring and impeller



Oversized wear ring installed

When the wear on the impeller and impeller flange has become excessive, it is time to think about reconditioning these parts. We would stress that reconditioning should always be considered before ordering new parts.

Reconditioning can be carried out by any of the FRAMO Service offices; or onboard if the necessary equipment and knowledge is available.

The first step of this reconditioning is to machine the impeller and the impeller flange down to new dimensions. The new dimensions on the impeller and the impeller flange will follow the oversized wear rings.

When the impeller and the impeller flange both have been machined down to the new dimensions, the oversized wear rings can be installed.

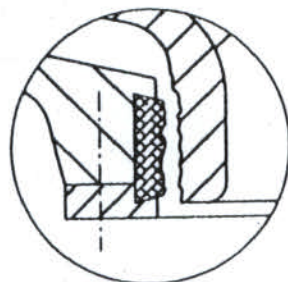
Oversize wear rings are carried on stock at all FRAMO Service offices.

We would point out that it is necessary to keep a very good record of which tanks/pumps that oversize wear rings have been installed in, to make sure that correct spares are kept and used for future repairs/maintenance.

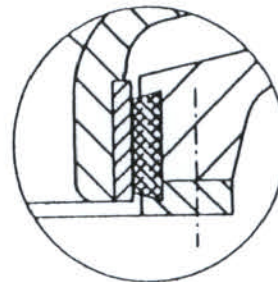
Then presumably oversize wear rings can be utilized quite a few times before again there is an excessive wear also on the impeller and impeller flange. Then comes the next step.

Step No. 3.

Back to original size wear rings



Worn-out oversized wear ring and impeller



Shrink-on ring together with original wear ring installed

When there is excessive wear on machined-down impeller and impeller flange, a shrink-on ring has to be fitted on these parts to bring the parts back to original size, allowing once more the use of original size wear rings.

The tolerances for remachining of the impeller and impeller flange to fit the standard shrink-on rings bringing the dimensions back to original tolerances will follow the shrink-on rings.

These shrink-on rings will also be carried on stock at all FRAMO Service offices.

As mentioned the remachining of impeller and impeller flange can normally be carried out by the crew onboard, provided of course that the vessel is equipped for doing this job. It is however very important to remember the alignment tolerances before machining - this to avoid unbalance in the cargopump rotary group.

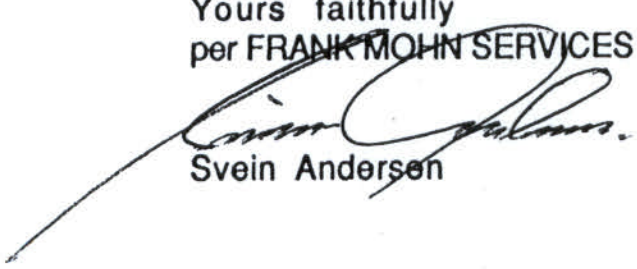
Should you have any doubt whether this can be carried out onboard your ship, we remind you that all FRAMO Service offices can do this job for you.

We further remind you again that a very close record has to be kept of which pumps/tanks that have oversize wear rings installed and which that have original size wear rings installed. Otherwise great problems can be the result if incorrect wear rings are installed during overhaul.

Hopefully this proposal will reduce your maintenance costs and contribute to improving the cargo pumping efficiency onboard your ships.

Good luck !

Yours faithfully
per FRANK MOHN SERVICES AS



Svein Andersen